Set 9 practice: Curl, divergence and line integrals

Zill §9.7, 9.8, 9.9

Quiz prep. Closed book, about 10 minutes per problem. Problems 1 and 2 are about div, curl and the conservative tests. Problems 3 to 5 are line integrals computed by parameterizing, for a scalar function, for work, and in the form $P\,dx + Q\,dy + R\,dz$. Problems 6 and 7 find a potential and use it, including a closed curve. Problem 8 is a field that is not conservative, so the only way through is to parameterize. No figures: every curve is described in words, with its parameterization and orientation stated. Every answer was recomputed independently by direct parameterization and, where a potential exists, through the potential.

Key methods

Divergence and curl. For $\mathbf{F} = \langle P, Q, R\rangle$:

$$\operatorname{div}\mathbf{F} = P_x + Q_y + R_z, \qquad \operatorname{curl}\mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ P & Q & R \end{vmatrix} = \langle R_y - Q_z,\ P_z - R_x,\ Q_x - P_y\rangle$$

Line integrals. With C given by $\mathbf{r}(t)$, $a \le t \le b$, and $ds = |\mathbf{r}'(t)|\,dt$:

$$\int_C f\,ds = \int_a^b f(\mathbf{r}(t))\,|\mathbf{r}'(t)|\,dt, \qquad \int_C \mathbf{F}\cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)\,dt = \int_C P\,dx + Q\,dy + R\,dz$$

The ds integral does not care about direction. The work integral changes sign when the orientation is reversed.

Conservative test. In 3D, $\operatorname{curl}\mathbf{F} = \mathbf{0}$. In 2D, $M_y = N_x$ for $\mathbf{F} = \langle M, N\rangle$. Either test settles it only on an open, simply connected domain (no holes where the field fails to exist).

Potential. Solve $\varphi_x = P$ by integrating in x (add an unknown function of the other variables), then match $\varphi_y = Q$ and $\varphi_z = R$ to pin that function down.

Fundamental theorem for line integrals. If $\mathbf{F} = \nabla\varphi$, then $\displaystyle\int_C \mathbf{F}\cdot d\mathbf{r} = \varphi(B) - \varphi(A)$, with A the start and B the end of C. The path does not matter, and a closed curve gives 0.

Each problem is on its own page with space for your working; the solution is on the next page.

Problem P1

Let $\mathbf{F} = \langle x^2y,\ yz,\ xz^2 \rangle$ and $\mathbf{G} = \langle 2xyz,\ x^2z,\ x^2y \rangle$.

(a) Find $\operatorname{div}\mathbf{F}$ and $\operatorname{curl}\mathbf{F}$, and check that $\operatorname{div}(\operatorname{curl}\mathbf{F}) = 0$.

(b) Is $\mathbf{G}$ conservative? If so, find a potential by inspection.

(c) Is $\mathbf{F}$ conservative?

Solution P1

Div, curl, and the 3D test

(a) Here $P = x^2y$, $Q = yz$, $R = xz^2$.

$$\operatorname{div}\mathbf{F} = 2xy + z + 2xz$$

For the curl, take the three components in turn:

$$R_y - Q_z = 0 - y = -y, \qquad P_z - R_x = 0 - z^2 = -z^2, \qquad Q_x - P_y = 0 - x^2 = -x^2$$

$$\boxed{\operatorname{div}\mathbf{F} = 2xy + z + 2xz, \qquad \operatorname{curl}\mathbf{F} = \langle -y,\ -z^2,\ -x^2\rangle}$$

The check: $\operatorname{div}\langle -y, -z^2, -x^2\rangle = 0 + 0 + 0 = 0$. This holds for the curl of any field with continuous second partials, so it catches arithmetic slips in the curl.

(b) For $\mathbf{G}$, $P = 2xyz$, $Q = x^2z$, $R = x^2y$:

$$R_y - Q_z = x^2 - x^2 = 0, \qquad P_z - R_x = 2xy - 2xy = 0, \qquad Q_x - P_y = 2xz - 2xz = 0$$

$\operatorname{curl}\mathbf{G} = \mathbf{0}$ and $\mathbf{G}$ is defined and smooth on all of $\mathbb{R}^3$, which is simply connected, so $\mathbf{G}$ is conservative. Looking at the first component, $2xyz$ is $\partial_x(x^2yz)$, and then $\partial_y(x^2yz) = x^2z$ and $\partial_z(x^2yz) = x^2y$ both match.

$$\boxed{\mathbf{G} \text{ is conservative, with potential } \varphi = x^2yz + C}$$

(c) $\operatorname{curl}\mathbf{F} = \langle -y, -z^2, -x^2\rangle$ is not the zero vector. At $(1, 1, 1)$ it is $\langle -1, -1, -1\rangle$. One point is enough.

$$\boxed{\mathbf{F} \text{ is not conservative}}$$

Problem P2

Decide which of these 2D fields are conservative. When the answer is yes, find a potential. When it is no, give a closed curve that proves it.

(a) $\mathbf{F} = \langle e^x \sin y + 2x,\ e^x \cos y \rangle$

(b) $\mathbf{F} = \langle y^2,\ x^2 \rangle$

(c) $\mathbf{F} = \left\langle \dfrac{-y}{x^2+y^2},\ \dfrac{x}{x^2+y^2} \right\rangle$ on the domain $\mathbb{R}^2$ minus the origin

Solution P2

$M_y = N_x$, and when it is not enough

(a) $M = e^x \sin y + 2x$ and $N = e^x \cos y$, defined on all of $\mathbb{R}^2$.

$$M_y = e^x \cos y, \qquad N_x = e^x \cos y$$

Equal on a simply connected domain, so conservative. Integrate $\varphi_x = M$ with respect to x:

$$\varphi = e^x \sin y + x^2 + g(y)$$

Then $\varphi_y = e^x \cos y + g'(y)$ must equal $N = e^x \cos y$, so $g' = 0$.

$$\boxed{\text{(a) conservative, } \varphi = e^x \sin y + x^2}$$

(b) $M = y^2$, $N = x^2$: $M_y = 2y$ and $N_x = 2x$. These differ wherever $y \neq x$.

$$\boxed{\text{(b) not conservative}}$$

(c) Compute both partials with the quotient rule, writing $r^2 = x^2 + y^2$:

$$M_y = \frac{-r^2 + 2y^2}{r^4} = \frac{y^2 - x^2}{r^4}, \qquad N_x = \frac{r^2 - 2x^2}{r^4} = \frac{y^2 - x^2}{r^4}$$

They are equal, but the test is not enough here. The domain is $\mathbb{R}^2$ with the origin removed, which has a hole, so $M_y = N_x$ no longer guarantees a potential. Test the unit circle, oriented counterclockwise, $\mathbf{r}(t) = \langle \cos t, \sin t \rangle$, $0 \leq t \leq 2\pi$. On it $r^2 = 1$, so $\mathbf{F}(\mathbf{r}(t)) = \langle -\sin t, \cos t \rangle$ and $\mathbf{r}'(t) = \langle -\sin t, \cos t \rangle$:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} (\sin^2 t + \cos^2 t)\, dt = 2\pi \neq 0$$

A conservative field gives 0 on every closed curve.

$$\boxed{\text{(c) not conservative: } \oint_C \mathbf{F} \cdot d\mathbf{r} = 2\pi \text{ on the unit circle, counterclockwise}}$$

Problem P3

Evaluate $\int_C f\,ds$.

(a) $f(x,y) = x$, and C is the part of the parabola $y = x^2$ from $(0,0)$ to $(1,1)$.

(b) $f(x,y,z) = x^2 + y^2 + z^2$, and C is the helix $\mathbf{r}(t) = \langle \cos t,\ \sin t,\ t \rangle$, $0 \le t \le 2\pi$.

Solution P3

Scalar line integrals, $ds = |\mathbf{r}'|\,dt$

(a) Parameterize with $x = t$: $\mathbf{r}(t) = \langle t, t^2\rangle$, $0 \le t \le 1$. Then $\mathbf{r}' = \langle 1, 2t\rangle$ and $ds = \sqrt{1+4t^2}\,dt$. On C, $f = x = t$.

$$\int_C x\,ds = \int_0^1 t\sqrt{1+4t^2}\,dt$$

Let $u = 1 + 4t^2$, so $du = 8t\,dt$, with u running from 1 to 5:

$$\int_0^1 t\sqrt{1+4t^2}\,dt = \frac{1}{8}\int_1^5 u^{1/2}\,du = \frac{1}{8}\cdot\frac{2}{3}\Big[u^{3/2}\Big]_1^5 = \frac{1}{12}(5\sqrt{5}-1)$$

$$\boxed{\int_C x\,ds = \frac{5\sqrt{5}-1}{12} \approx 0.8484}$$

(b) $\mathbf{r}'(t) = \langle -\sin t,\ \cos t,\ 1\rangle$, so $|\mathbf{r}'| = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{2}$, a constant. On C, $x^2 + y^2 + z^2 = \cos^2 t + \sin^2 t + t^2 = 1 + t^2$.

$$\int_C (x^2+y^2+z^2)\,ds = \sqrt{2}\int_0^{2\pi}(1+t^2)\,dt = \sqrt{2}\Big(2\pi + \frac{8\pi^3}{3}\Big)$$

$$\boxed{\int_C (x^2+y^2+z^2)\,ds = \sqrt{2}\Big(2\pi + \frac{8\pi^3}{3}\Big) = \frac{2\sqrt{2}\,\pi\,(3+4\pi^2)}{3} \approx 125.8}$$

Running the helix backward gives the same number, because $ds = |\mathbf{r}'|\,dt$ is always positive.

Problem P4

Find the work done by $\mathbf{F} = \langle y,\ x^2 \rangle$ along the upper half of the circle $x^2 + y^2 = 4$, from $(2, 0)$ to $(-2, 0)$. Then say what the work is for the opposite direction.

Solution P4

Work, orientation stated first

The path leaves $(2,0)$, passes over $(0,2)$ and arrives at $(-2,0)$, which is counterclockwise. Parameterize with that orientation:

$$\mathbf{r}(t) = \langle 2\cos t,\ 2\sin t\rangle, \qquad 0 \le t \le \pi$$

At $t = 0$ this is $(2,0)$ and at $t = \pi$ it is $(-2,0)$, so increasing t matches the stated direction. Then $\mathbf{r}'(t) = \langle -2\sin t,\ 2\cos t\rangle$ and $\mathbf{F}(\mathbf{r}(t)) = \langle 2\sin t,\ 4\cos^2 t\rangle$.

$$\mathbf{F}\cdot\mathbf{r}' = -4\sin^2 t + 8\cos^3 t$$

$$W = \int_0^{\pi} \left(-4\sin^2 t + 8\cos^3 t\right) dt$$

Use $\sin^2 t = \frac{1}{2}(1-\cos 2t)$, which gives $\int_0^{\pi}\sin^2 t\,dt = \frac{\pi}{2}$. For the cubic, $\cos^3 t = (1-\sin^2 t)\cos t$ has antiderivative $\sin t - \frac{1}{3}\sin^3 t$, which is 0 at both 0 and π. So

$$W = -4\cdot\frac{\pi}{2} + 8\cdot 0 = -2\pi$$

$$\boxed{W = -2\pi \text{ for the counterclockwise path from } (2,0) \text{ to } (-2,0)}$$

Going from $(-2,0)$ to $(2,0)$ over the same arc reverses the orientation and flips the sign: $W = +2\pi$. Check: $\mathbf{r}(t) = \langle 2\cos(\pi - t),\ 2\sin(\pi - t)\rangle$ on $0 \le t \le \pi$ gives exactly $+2\pi$.

Problem P5

Evaluate $\int_C y\,dx + z\,dy + x\,dz$, where C is the twisted cubic $x = t,\ y = t^2,\ z = t^3$, traced from $(0, 0, 0)$ to $(1, 1, 1)$ as t goes from 0 to 1. Is there a shortcut?

Solution P5

The $P\,dx + Q\,dy + R\,dz$ form

The integral is the work of $\mathbf{F} = \langle y, z, x\rangle$ along C. Substitute $x = t$, $y = t^2$, $z = t^3$, along with $dx = dt$, $dy = 2t\,dt$, $dz = 3t^2\,dt$:

$$y\,dx = t^2\,dt, \qquad z\,dy = t^3 \cdot 2t\,dt = 2t^4\,dt, \qquad x\,dz = t \cdot 3t^2\,dt = 3t^3\,dt$$

$$\int_C y\,dx + z\,dy + x\,dz = \int_0^1 \left(t^2 + 3t^3 + 2t^4\right) dt = \frac{1}{3} + \frac{3}{4} + \frac{2}{5} = \frac{20 + 45 + 24}{60}$$

$$\boxed{\int_C y\,dx + z\,dy + x\,dz = \frac{89}{60}}$$

No shortcut: $\operatorname{curl}\langle y, z, x\rangle = \langle R_y - Q_z,\ P_z - R_x,\ Q_x - P_y\rangle = \langle -1, -1, -1\rangle \neq \mathbf{0}$, so the field is not conservative and the value depends on the path. Another curve from $(0, 0, 0)$ to $(1, 1, 1)$, such as the straight segment, would give a different number. Traced from $(1, 1, 1)$ back to the origin the same curve gives $-\frac{89}{60}$.

Problem P6

Let $\mathbf{F} = \langle 2xyz,\ x^2z + e^z,\ x^2y + ye^z \rangle$. Show that $\mathbf{F}$ is conservative, find a potential, and use it to find $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$, where C is the helix $\mathbf{r}(t) = \langle \cos t,\ \sin t,\ t \rangle$ from $t = 0$ to $t = \pi/2$.

Solution P6

Potential function in 3D, then the fundamental theorem

Test. $P = 2xyz$, $Q = x^2z + e^z$, $R = x^2y + ye^z$.

$$R_y - Q_z = (x^2 + e^z) - (x^2 + e^z) = 0, \qquad P_z - R_x = 2xy - 2xy = 0, \qquad Q_x - P_y = 2xz - 2xz = 0$$

The curl is zero on all of $\mathbb{R}^3$, so $\mathbf{F}$ is conservative.

Potential. Integrate $\varphi_x = 2xyz$ in x:

$$\varphi = x^2yz + g(y, z)$$

Match $\varphi_y = x^2z + g_y$ to $Q = x^2z + e^z$, so $g_y = e^z$ and $g = ye^z + h(z)$. Now

$$\varphi = x^2yz + ye^z + h(z)$$

Match $\varphi_z = x^2y + ye^z + h'(z)$ to $R = x^2y + ye^z$, so $h' = 0$.

$$\varphi = x^2yz + ye^z$$

Evaluate. The helix starts at $\mathbf{r}(0) = (1, 0, 0)$ and ends at $\mathbf{r}(\pi/2) = (0, 1, \pi/2)$, with increasing t as the orientation.

$$\varphi(1, 0, 0) = 0 + 0 = 0, \qquad \varphi\big(0, 1, \tfrac{\pi}{2}\big) = 0 + 1 \cdot e^{\pi/2} = e^{\pi/2}$$

$$\boxed{\varphi = x^2yz + ye^z, \qquad \int_C \mathbf{F} \cdot d\mathbf{r} = \varphi(B) - \varphi(A) = e^{\pi/2} \approx 4.810}$$

Parameterizing the helix directly would need $\int_0^{\pi/2}$ of a sum of products of trigonometric functions with t and e^t. The theorem skips all of it, and any other curve from $(1, 0, 0)$ to $(0, 1, \pi/2)$ gives the same value.

Problem P7

Let $\mathbf{F} = \langle 3x^2y^2 - \cos y, \quad 2x^3y + x\sin y \rangle$.

(a) Show that $\mathbf{F}$ is conservative and find a potential.

(b) Evaluate $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$, where C is the curve $\mathbf{r}(t) = \langle t^2, \ \pi t \rangle$, $0 \le t \le 1$, oriented in the direction of increasing t.

(c) Evaluate $\displaystyle\oint_{C'} \mathbf{F} \cdot d\mathbf{r}$, where C' is the circle of radius 2 centered at the origin, traversed once counter-clockwise.

Solution P7

2D potential, open path, closed path

(a) $M = 3x^2y^2 - \cos y$ and $N = 2x^3y + x\sin y$ are defined on all of $\mathbb{R}^2$.

$$M_y = 6x^2y + \sin y, \qquad N_x = 6x^2y + \sin y$$

Equal on a simply connected domain, so $\mathbf{F}$ is conservative. Integrate $\varphi_x = M$:

$$\varphi = x^3y^2 - x\cos y + g(y)$$

Then $\varphi_y = 2x^3y + x\sin y + g'(y)$ matches N when $g' = 0$.

$$\varphi = x^3y^2 - x\cos y$$

(b) The curve starts at $\mathbf{r}(0) = (0,0)$ and ends at $\mathbf{r}(1) = (1,\pi)$. It is an arc of the parabola $x = y^2/\pi^2$. Its shape does not matter, only the endpoints:

$$\varphi(1,\pi) = \pi^2 - \cos\pi = \pi^2 + 1, \qquad \varphi(0,0) = 0 - 0 = 0$$

$$\boxed{\int_C \mathbf{F}\cdot d\mathbf{r} = \varphi(1,\pi) - \varphi(0,0) = \pi^2 + 1 \approx 10.87}$$

(c) A closed curve starts and ends at the same point A, so $\varphi(B) - \varphi(A) = 0$. The circle lies in the domain $\mathbb{R}^2$ where $\mathbf{F}$ is smooth, so the theorem applies.

$$\boxed{\oint_{C'} \mathbf{F}\cdot d\mathbf{r} = 0}$$

Contrast this with problem 2(c). There the field was undefined at the origin, which the circle encloses, so the closed-curve argument failed. Here nothing is missing inside the circle.

Problem P8

Let $\mathbf{F} = \langle xy,\ x^2 \rangle$. Test whether $\mathbf{F}$ is conservative. Then find the work from $(0,0)$ to $(1,1)$ along

(a) C_1: the straight line $y = x$, and

(b) C_2: the parabola $y = x^2$,

both oriented from $(0,0)$ toward $(1,1)$. Finally, find $\oint \mathbf{F} \cdot d\mathbf{r}$ for the closed curve that follows C_2 out to $(1,1)$ and returns along C_1 reversed.

Solution P8

Not conservative, so parameterize every path

Test. $M = xy$ and $N = x^2$. $M_y = x$ and $N_x = 2x$. These differ for every $x \neq 0$, so $\mathbf{F}$ is not conservative. No potential exists, so the endpoints alone do not determine the work and each path has to be parameterized.

(a) C_1: $\mathbf{r}(t) = \langle t, t\rangle$, $0 \le t \le 1$. Then $dx = dt$, $dy = dt$ and $\mathbf{F}(\mathbf{r}(t)) = \langle t^2, t^2\rangle$.

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_0^1 (t^2 + t^2)\, dt = \int_0^1 2t^2\, dt = \frac{2}{3}$$

(b) C_2: $\mathbf{r}(t) = \langle t, t^2\rangle$, $0 \le t \le 1$. Then $dx = dt$, $dy = 2t\, dt$ and $\mathbf{F}(\mathbf{r}(t)) = \langle t^3, t^2\rangle$.

$$\int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_0^1 (t^3 + t^2 \cdot 2t)\, dt = \int_0^1 3t^3\, dt = \frac{3}{4}$$

The two paths share endpoints and disagree, which confirms the test: $\frac{2}{3} \neq \frac{3}{4}$.

The closed curve is C_2 followed by $-C_1$, and reversing a path negates the integral:

$$\oint \mathbf{F} \cdot d\mathbf{r} = \int_{C_2} \mathbf{F} \cdot d\mathbf{r} - \int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \frac{3}{4} - \frac{2}{3} = \frac{1}{12}$$

$$\boxed{\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \frac{2}{3}, \qquad \int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \frac{3}{4}, \qquad \oint \mathbf{F} \cdot d\mathbf{r} = \frac{1}{12} \neq 0}$$

A conservative field would give 0 for the loop. A third route, along the x-axis to $(1, 0)$ and then straight up to $(1, 1)$, gives $0 + 1 = 1$: along the axis $y = 0$ and $dy = 0$ so $\mathbf{F} \cdot d\mathbf{r} = 0$, and up the line $x = 1$ the integrand is $x^2\, dy = dy$. That is a third value for the same endpoints.